

netan

netan is a Rust-first local network analysis tool for OSM-based routing research.

This repository now includes:

- a shared Rust routing core
- a CLI for reproducible local runs
- a functional native egui desktop shell
- an in-repo QGIS plugin for QGIS 3 and QGIS 4 that drives the CLI and loads spatial outputs
- explicit manifests and bundles under `.netan/`
- local exports to json, csv, geojson, gpkg, parquet, and geoparquet

Via-way turn restriction handling is present. Turn penalties, acceleration structures, and broader packaging are still pending. See [PROGRESS.md](#).

Workspace Layout

- `crates/cli`: netan CLI entry point
- `crates/gui`: native desktop shell
- `examples/profiles`: reusable example profiles
- `examples/requests`: route, OD, and matrix inputs
- `qgis_plugin/netan_qgis`: QGIS plugin package

Build

```
cargo fmt --all
cargo check
cargo test
```

CLI Quickstart

Import a dataset:

```
cargo run -p netan-cli -- dataset import datasets/
groningen-260317.osm.pbf --name groningen_2026_03
```

Validate and compile a profile:

```
cargo run -p netan-cli -- profile validate examples/profiles/
car_research_v1.yml
cargo run -p netan-cli -- profile compile --dataset
groningen_2026_03 --profile examples/profiles/car_research_v1.yml
```

Run a route and write a spatial result QGIS can open directly:

```
cargo run -p netan-cli -- analyze route \
--dataset groningen_2026_03 \
--profile examples/profiles/car_research_v1.yml \
--request examples/requests/route.json \
--out .netan/runs/example-route.geojson
```

Run OD and matrix examples:

```
cargo run -p netan-cli -- analyze od \
--dataset groningen_2026_03 \
--profile examples/profiles/car_research_v1.yml \
--pairs examples/requests/od_pairs.csv \
--out .netan/runs/example-od.gpkg

cargo run -p netan-cli -- analyze matrix \
--dataset groningen_2026_03 \
--profile examples/profiles/car_research_v1.yml \
--origins examples/requests/matrix_origins.csv \
--destinations examples/requests/matrix_destinations.csv \
--out .netan/runs/example-matrix.gpkg
```

Inspect persistent state:

```
cargo run -p netan-cli -- cache list
```

Native GUI

Launch the desktop app:

```
cargo run -p netan-cli -- gui
```

Recommended GUI flow:

1. In Datasets, import datasets/groningen-260317.osm.pbf as groningen_2026_03.
2. In Profiles, validate examples/profiles/car_research_v1.yml and compile it for the selected dataset.
3. In Analyses, use the example request files or write a new route request from typed coordinates.
4. Set an output like .netan/runs/gui-route.geojson or .netan/runs/gui-matrix.gpkg.
5. Execute the analysis and inspect the result from Runs.

Notes:

- GUI paths can be repository-relative or absolute.
- The GUI auto-compiles the selected profile on demand if the needed metric bundle is missing.
- Spatial outputs are still written by the same report/export layer as the CLI.

QGIS Plugin

The QGIS plugin lives under `qgis_plugin/netan_qgis`. Full setup instructions are in `qgis_plugin/README.md`.

Minimal install flow:

1. Build the CLI binary:

```
cargo build -p netan-cli
```

1. Symlink the plugin into your QGIS plugin directory.

Linux QGIS example:

```
mkdir -p ~/.local/share/QGIS/QGIS3/profiles/default/python/plugins
ln -s /home/elmeriniemi/stuff/netan/qgis_plugin/netan_qgis
~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/
netan_qgis
```

Windows QGIS 4 example for a repo living in WSL:

```
New-Item -ItemType Directory -Force
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"
cmd /c mklink /D
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"
"\\wsl$
\Ubuntu\home\elmeriniemi\stuff\netan\qgis_plugin\netan_qgis"
```

1. In QGIS, set either native or WSL mode.

Native mode:

- Execution mode: Native
- QGIS-visible workspace root: `/home/elmeriniemi/stuff/netan`
- Native netan executable: `/home/elmeriniemi/stuff/netan/target/debug/netan`
- Profile: `examples/profiles/car_research_v1.yml`

Windows QGIS with WSL-built netan:

- Execution mode: WSL
- QGIS-visible workspace root: `\\wsl$
\Ubuntu\home\elmeriniemi\stuff\netan`

- WSL distro: Ubuntu
- WSL workspace root: /home/elmeriniemi/stuff/netan
- WSL netan executable: /home/elmeriniemi/stuff/netan/target/debug/netan
- Profile: examples/profiles/car_research_v1.yml

1. Press Refresh, choose dataset groningen_2026_03, then use:

- Run Route with output .netan/runs/qgis-route.geojson
- Run OD with output .netan/runs/qgis-od.gpkg
- Run Matrix with output .netan/runs/qgis-matrix.gpkg

The plugin auto-loads spatial outputs into the current QGIS project after successful runs, including when Windows QGIS is talking to a WSL-hosted workspace through `\\wsl$`.

Full Windows Example

This section assumes:

- your source repo lives in WSL at /home/elmeriniemi/stuff/netan
- you have a Windows-visible mount or symlink at /windownloads
- you want to use Windows QGIS 4

Option A: Use Windows QGIS 4 with the WSL build

This is the simplest path if you are already developing in WSL.

1. Build netan in WSL:

```
cd /home/elmeriniemi/stuff/netan
cargo build -p netan-cli
```

1. Make sure the workspace has the data and outputs you want available to Windows QGIS.

If you want to copy the repo to Windows but skip Rust build output:

```
cd /home/elmeriniemi/stuff/netan
mkdir -p /windownloads/netan
for p in .* *; do
    [ "$p" = "." ] || [ "$p" = ".." ] || [ "$p" = "target" ] || cp -
a -- "$p" /windownloads/netan/
done
```

That keeps original data such as datasets/, examples/, and qgis_plugin/.

1. Add the QGIS 4 plugin from Windows PowerShell.

If you want QGIS to use the plugin directly from the WSL repo:

```
New-Item -ItemType Directory -Force
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"
cmd /c mklink /D
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"
"\\wsl$
\Ubuntu\home\elmeriniemi\stuff\netan\qgis_plugin\netan_qgis"
```

If you copied the repo to Windows first, point the symlink at the Windows copy instead:

```
New-Item -ItemType Directory -Force
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"
cmd /c mklink /D
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"
"C:\path\to\netan\qgis_plugin\netan_qgis"
```

1. Start QGIS 4 and enable the plugin from Plugins -> Manage and Install Plugins.
2. In the plugin dock, use these settings for WSL-backed execution:

- Execution mode: WSL
- QGIS-visible workspace root: \\wsl\$\Ubuntu\home\elmeriniemi\stuff\netan
- WSL distro: Ubuntu
- WSL workspace root: /home/elmeriniemi/stuff/netan
- WSL netan executable: /home/elmeriniemi/stuff/netan/target/debug/netan
- Profile: examples/profiles/car_research_v1.yml

1. Press Refresh, choose dataset groningen_2026_03, then run:

- Route with output .netan/runs/qgis-route.geojson
- OD with output .netan/runs/qgis-od.gpkg
- Matrix with output .netan/runs/qgis-matrix.gpkg

Option B: Use a native Windows build

Use this if you want QGIS to run netan.exe directly without wsl.exe.

1. Copy the repo to Windows.

From WSL:

```
cd /home/elmeriniemi/stuff/netan
mkdir -p /windowdownloads/netan
for p in .* *; do
    [ "$p" = "." ] || [ "$p" = ".." ] || [ "$p" = "target" ] || cp -
```

```
a -- "$p" /windownloads/netan/  
done
```

1. Open Windows PowerShell, then build the CLI from the Windows copy:

```
cd C:\path\to\netan  
cargo build -p netan-cli
```

1. Install the plugin into QGIS 4:

```
New-Item -ItemType Directory -Force  
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins"  
cmd /c mklink /D  
"$env:APPDATA\QGIS\QGIS4\profiles\default\python\plugins\netan_qgis"  
"C:\path\to\netan\qgis_plugin\netan_qgis"
```

1. In QGIS 4, use:

- Execution mode: Native
- QGIS-visible workspace root: C:\path\to\netan
- Native netan executable: C:
 \path\to\netan\target\debug\netan.exe
- Profile: examples/profiles/car_research_v1.yml

1. Import and run from PowerShell or from the native GUI/CLI:

```
cd C:\path\to\netan  
.\target\debug\netan.exe dataset import  
datasets\ groningen-260317.osm.pbf --name groningen_2026_03  
.\target\debug\netan.exe profile compile --dataset  
 groningen_2026_03 --profile examples\profiles\car_research_v1.yml  
.\target\debug\netan.exe analyze route --dataset  
 groningen_2026_03 --profile examples\profiles\car_research_v1.yml  
--request examples\requests\route.json --out .netan\runs\example-  
route.geojson
```

1. In the plugin, press Refresh and run analyses normally.

Which Windows setup should you use?

- Use WSL mode if your active repo, datasets, and build artifacts live in WSL.
- Use Native mode if you want a pure Windows setup and are willing to build netan.exe from the Windows copy.

Input Examples

Route request JSON:

```
{
  "route_id": "route_demo_001",
  "origin": { "id": "origin_a", "lon": 6.5665, "lat": 53.2194 },
  "destination": { "id": "destination_b", "lon": 6.5716, "lat":
53.2148 },
  "snap": { "max_distance_m": 500.0 },
  "returns": { "geometry": "full", "segment_rows": true }
}
```

OD CSV:

```
id,source_x,source_y,target_x,target_y
od_demo_001,6.5665,53.2194,6.5716,53.2148
od_demo_002,6.5636,53.2181,6.5698,53.2137
```

Matrix point-set CSV:

```
id,x,y
origin_a,6.5665,53.2194
origin_c,6.5636,53.2181
```